INCH-POUND

MIL-PRF-27/372
27 April 2012

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE),
INDUCTOR, SURFACE MOUNT, LEADLESS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

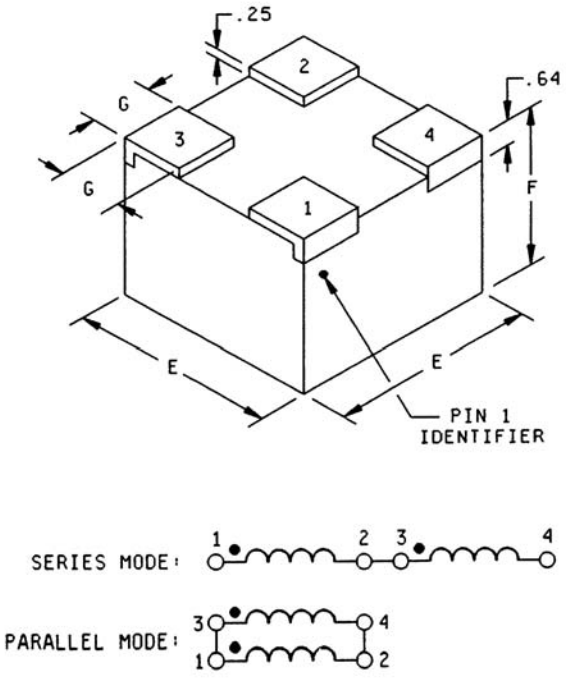


FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5950

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Size	E_{max} Millimeters (inches)	F_{max} Millimeters (inches)	G 1/ Millimeters (inches)
A	6.60 (.260)	5.84 (.230)	2.16 (.085)
B	10.41 (.410)	5.84 (.230)	3.43 (.135)
C	10.41 (.410)	7.87 (.310)	3.43 (.135)
D	13.59 (.535)	7.87 (.310)	5.08 (.200)

1/ The tolerance for all dimensions shall be ±0.25 mm (.010 inch).

mm	Inches
0.25	.010
0.64	.025

NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

FIGURE 1. Dimensions and configurations - continued.

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TABLE I. Electrical characteristics.

Dash numbers	Series			Parallel			Configur-ation	Power rating (mW)
	Inductance ±10 percent (mH)	Rated dc current (mA)	DC resistance (max)	Inductance ±10 percent (mH)	Rated dc current (mA)	DC resistance (max)		
001	0.100	300	1.60	0.025	600	0.400	A	125
002	0.100	600	0.96	0.025	1200	0.240	B	250
003	0.100	780	0.37	0.025	1560	0.093	C	300
004	0.100	1350	0.21	0.025	2700	0.053	D	450
005	0.250	200	3.11	0.062	400	0.778	A	125
006	0.250	400	1.98	0.062	800	0.495	B	250
007	0.250	530	0.82	0.062	1060	0.205	C	300
008	0.250	930	0.41	0.062	1860	0.103	D	450
009	0.500	150	6.14	0.125	300	1.535	A	125
010	0.500	300	3.35	0.125	600	0.838	B	250
011	0.500	380	1.62	0.125	760	0.405	C	300
012	0.500	660	0.82	0.125	1320	0.205	D	450
013	0.750	120	9.45	0.187	240	2.363	A	125
014	0.750	250	5.20	0.187	500	1.300	B	250
015	0.750	310	2.50	0.187	620	0.625	C	300
016	0.750	540	1.25	0.187	1080	0.313	D	450
017	1.000	95	12.85	0.250	190	3.213	A	125
018	1.000	200	6.90	0.250	400	1.725	B	250
019	1.000	270	3.37	0.250	540	0.843	C	300
020	1.000	470	1.87	0.250	940	0.468	D	450
021	2.000	75	25.60	0.500	150	6.400	A	125
022	2.000	150	12.20	0.500	300	3.050	B	250
023	2.000	190	6.75	0.500	380	1.690	C	300
024	2.000	330	3.62	0.500	660	0.910	D	450
025	3.000	60	38.40	0.750	120	9.600	A	125
026	3.000	120	18.30	0.750	240	4.580	B	250
027	3.000	160	10.12	0.750	320	2.530	C	300
028	3.000	270	5.50	0.750	540	1.380	D	450
029	5.000	45	64.00	1.250	90	16.000	A	125
030	5.000	90	30.40	1.250	180	7.600	B	250
031	5.000	120	16.87	1.250	240	4.220	C	300
032	5.000	210	9.12	1.250	420	2.280	D	450
033	7.500	30	96.00	1.875	60	24.000	A	125
034	7.500	70	46.00	1.875	140	11.500	B	250
035	7.500	100	25.37	1.875	200	6.350	C	300
036	7.500	170	13.75	1.875	340	3.440	D	450
037	10.000	20	128.00	2.500	40	32.000	A	125
038	10.000	50	61.00	2.500	100	15.250	B	250
039	10.000	80	33.75	2.500	160	8.440	C	300
040	10.000	140	18.25	2.500	280	4.570	D	450

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Design and construction:

Dimensions and configurations: See figure 1.

Case: Grade 5, encapsulated.

Terminals: The terminals shall be tin-lead plated phosphor bronze.

Weight: 1.25 ounces, maximum.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +125°C.

Resistance to soldering heat: MIL-STD-202, method 210 test condition B. Depth of immersion shall be the entire mounting surface for 4 to 5 seconds.

Bond Strength: MIL-STD-883, method 2011, test condition F.

Force: 2 pounds.

Barometric pressure: MIL-STD-202, method 105, test condition C, (70,000 feet), test voltage 100 V rms.

Insulation resistance: MIL-STD-202, method 302, 1,000 megohms minimum at 100 V dc.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 200 volts rms.

At reduced barometric pressure: 100 volts rms.

Vibration (high frequency): MIL-STD-202, method 204, test condition D.

Moisture resistance: MIL-STD-202, method 106, method of mounting shall be on a test substrate that provide test pieces on inch minimum separation.

Temperature rise: The temperature rise shall be 35°C over ambient temperature of 90°C.

Marking location: Marking shall be on top of the case.

Part or Identifying Number (PIN): M27/372-(dash number from table I).

Qualification: Qualification testing and approval to M27/372-040 shall be sufficient to grant qualification approval to M27/372-001 through M27/372-040 inclusive.

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202 MIL-STD-883

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2011-005)

Review activities:
Army – AR, MI
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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